

Notes: Ericson and Pakes (RES, 1995)

Markov-Perfect Industry Dynamics: A Framework for Empirical Work

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1 Big picture

- **Question.** How can we build a dynamic industry model with firm heterogeneity, idiosyncratic uncertainty, endogenous entry and exit, and strategic interaction that is rich enough to organize empirical work?
- **Setting.** Firms invest to explore and develop profit opportunities in an evolving market. Their own investment success pushes them toward more profitable states, while rivals' successes and outside competition push them toward less profitable states.
- **Core challenge.** Standard industry models often generate either deterministic long-run outcomes, representative firms, or no meaningful firm turnover. They therefore miss the simultaneous entry and exit, persistent heterogeneity, and rank reversals seen in firm-level data.
- **Main modeling contribution.** The paper sets up a dynamic game in which each firm's state is its current *success level*, the industry state is the cross-sectional distribution of firms across success states, and optimal investment, entry, and exit jointly determine the law of motion for industry structure.
- **Main theoretical result.** A firm's problem has a unique bounded value function and well-defined investment and exit rules. There are finite no-investment and exit boundaries, each firm dies in finite time with probability one, entry is finite, and a rational-expectations Markov-perfect equilibrium exists.
- **Main dynamic result.** The equilibrium industry process is **ergodic**. The industry does not converge to a single permanent structure. Instead, it cycles through a recurrent class of structures with a unique invariant distribution.
- **Main quantitative lesson from the example.** In the homogeneous-product Cournot example, the Markov-perfect Nash equilibrium is usually a duopoly but exhibits frequent entry and exit. Relative to a planner or a colluder, it has more firms, more investment, lower prices, and intermediate welfare.
- **Why this paper matters.** This is a foundational framework for empirical dynamic industrial organization. It is the intellectual starting point for later work on dynamic games, firm turnover, industry evolution, and structural policy counterfactuals.

2 Data and measurement (key objects)

2.1 Firm-level state and decisions in the model

- **Firm state.** Each incumbent firm is indexed by a scalar success state $\omega \in \mathbb{Z}$. A higher ω means the firm is in a stronger competitive position relative to both rivals inside the industry and alternatives outside it.
- **Firm decisions.** In each period, the firm chooses whether to continue or exit and, if it continues, how much to invest, $x \in \mathbb{R}_+$.
- **Current payoff.** A reduced-form spot-market equilibrium gives operating profits $A(\omega, s)$, where s is industry structure. Net current cash flow is

$$R(\omega, s; x) = A(\omega, s) - c(\omega)x.$$

- **Economic interpretation.** The model is about active exploration: investment does not simply scale current output; it changes the probability of moving to better future states.

2.2 Industry-level state

- **Industry structure.** The industry state is a counting measure

$$s = (s_\omega)_{\omega \in \Omega},$$

where s_ω is the number of firms currently at success state ω .

- **Entry states.** Potential entrants draw an initial state from a set $\Omega^e \subseteq \Omega$.
- **Competition ordering.** The paper assumes a complete preorder over structures, written informally as “stronger” or “weaker” competition. Profits are weakly decreasing in stronger competition and weakly increasing in own success ω .
- **State couple.** The pair (ω, s) summarizes everything that matters for a firm’s current and future payoffs.

2.3 Unobservables and timing

- **Own innovation shock.** The firm’s investment produces a stochastic own outcome, denoted below by an increment r , which can move the firm to a better state.
- **Outside-competition shock.** An exogenous process, denoted below by q , shifts all firms toward worse states. This captures things like stronger outside substitutes, adverse cost shifts, or outside technological progress.
- **Entrant quality.** Entrants do not know ex ante exactly how competitive they will be once set up. They only know a distribution over entry states.
- **Timing.** In each period, incumbents observe the current state (ω_t, s_t) , choose investment and exit, entrants decide whether to pay the sunk entry cost, shocks are realized, and then next period’s industry structure is formed.
- **Information structure.** Firms know the law of motion of the environment and their own transition probabilities, but they do not directly observe rivals’ current investment choices. This makes the equilibrium Markov-perfect with imperfectly observable actions.

2.4 Numerical illustration

- **Example environment.** The paper simulates a homogeneous-product Cournot industry with firm-specific marginal costs.
- **Comparison objects.** It computes the Markov-perfect Nash equilibrium, a multi-plant colluder’s optimum, and a social planner’s optimum under the same primitives.
- **Simulation design.** Descriptive statistics come from a long simulation starting with one entrant. Welfare is summarized from repeated finite-horizon runs.

3 Models and empirical specifications (all equations + interpretation)

3.1 Primitive objects and economic environment

The model primitives are the current-profit function $A(\omega, s)$, the investment-cost schedule $c(\omega)$, the firm's transition probabilities, the entrant-state distribution, the entry-cost schedule, and the discount factor $\beta \in (0, 1)$.

The firm's net current return from remaining active and investing x is

$$R(\omega, s; x) = A(\omega, s) - c(\omega)x.$$

- **Interpretation.** $A(\omega, s)$ is a reduced form for the spot-market equilibrium. The paper deliberately stays agnostic about whether competition is Cournot, Bertrand, or something else until the numerical example.
- **Monotonicity assumptions.** A firm with a higher success state ω is weakly more profitable, and stronger industry competition weakly lowers current profits.

3.2 Transition process

A firm's next-period state depends on its current state, its investment, and the outside environment. In the general model the transition probability is written as

$$p(\omega' \mid \omega, x).$$

The paper motivates this as the convolution of two forces:

- a firm-specific investment outcome that can move the firm upward,
- and an exogenous outside-competition shock that can move all firms downward.

In the example below, the law of motion takes the simple form

$$\omega' = \omega + r + q,$$

where r comes from own R&D success and $q \leq 0$ captures drift from outside competition.

Interpretation. This is the key engine of industry dynamics. Investment can improve a firm's position, but there is a permanent force pushing all firms toward obsolescence unless they keep innovating.

3.3 Incumbent firm's dynamic problem

An incumbent chooses continuation and investment to maximize the expected present value of net cash flows plus the payoff from exiting. Let $\chi_t \in \{0, 1\}$ be the continuation decision and let ϕ denote the payoff from exiting (the paper often discusses the exit region after normalizing this outside option). The firm's dynamic problem is represented recursively by

$$V(\omega, s) = \max \left\{ \sup_{x \geq 0} \left[R(\omega, s; x) + \beta \sum_q \sum_{\omega'} \sum_{s'} V(\omega', s') p(\omega' \mid \omega, x, q) q_\omega(s' \mid s, q) p_q \right], \phi \right\}.$$

- $q_\omega(s' | s, q)$ is the firm's perceived transition probability for competitors' states and the industry structure.
- The first term inside the max is the value of continuing and investing optimally.
- The second term is the value of terminating the project.

Interpretation. This is a standard Bellman equation, but the state variable is industry structure itself. That is what makes the framework an industry-dynamics model rather than a single-firm learning model.

3.4 Entrant problem and free-entry condition

Potential entrants are identical ex ante. To enter, they pay a sunk setup cost x_m^e , which can depend on the number of simultaneous entrants m . Entry takes one period, and successful entrants appear next period at a random entry state.

Let $V^e(s, m)$ denote the expected value of entering when the current structure is s and m firms enter. In reduced form,

$$V^e(s, m) = \mathbb{E}[V(\omega^e, s_{+1}) | s, m],$$

where the expectation integrates over entrant qualities, rivals' transitions, and the outside shock.

The number of entrants is determined by a free-entry rule:

$$m(s) = \min \{m \in \mathbb{Z}_+ : V^e(s, m) \geq x_m^e \text{ and } V^e(s, m+1) < x_{m+1}^e\},$$

with zero entry if even a single entrant would not find entry profitable.

Interpretation. Entry is endogenous, finite, and state dependent. Strong competition lowers the value of entry and eventually chokes it off.

3.5 Equilibrium definition

An equilibrium consists of the value function, optimal incumbent investment and exit rules, a law of motion for industry structure, and an entry rule, such that:

- each incumbent solves its Bellman problem given beliefs about industry transitions,
- entrants satisfy the free-entry condition,
- and those beliefs are correct in equilibrium because the optimal decisions of all incumbents and entrants aggregate into the actual transition kernel for industry structure.

Denoting the equilibrium transition kernel for structures by $Q(s' | s)$, the equilibrium is a **rational-expectations Markov-perfect Nash equilibrium**.

Interpretation. Firms behave optimally using a forecast of future industry states, and the forecast is exactly the one generated when every firm behaves that way.

3.6 Result 1: optimal incumbent behavior

The first main proposition establishes that the incumbent's problem is well behaved.

- There is a unique bounded value function $V(\omega, s)$, increasing in own success state ω .
- There is a unique bounded optimal investment rule $x(\omega, s)$.
- There is an optimal exit rule $\chi(\omega, s) \in \{0, 1\}$.

More importantly, the optimal policy has threshold structure:

- there is a **lower coasting region** in which the firm neither exits nor invests,
- an **upper coasting region** in which the firm is successful enough that it stops investing,
- and a **lower liquidation boundary** below which it exits.

Interpretation.

- **Upper coasting:** when a firm is already very successful, the marginal value of pushing even further up the state space becomes smaller than the marginal cost of additional investment.
- **Lower coasting:** a weak firm may temporarily stop investing but still remain alive because future improvements in the environment may make reactivation worthwhile.
- **Liquidation:** if the firm drifts too far down and outside pressure keeps building, it eventually exits.

A further implication is that the firm's lifetime is finite with probability one.

3.7 Result 2: finite entry and compact state space

The second result shows that as the number of active firms rises, the value of being in the industry can be made arbitrarily close to the exit value. Formally, fixing a state ω and adding more firms at that state,

$$\lim_{n \rightarrow \infty} V(\omega, s + ne_\omega) = \phi.$$

This implies two corollaries:

- there is an upper bound on the number of profitable entrants in any period,
- and there is an industry size large enough that additional entry stops altogether.

Interpretation. Entry cannot explode. Competition eventually becomes strong enough that further entry is no longer worth paying for. This endogenously makes the relevant structure space finite and computationally manageable.

3.8 Result 3: existence of equilibrium

Using the boundedness and compactness generated by the first two results, the paper proves existence of a rational-expectations equilibrium. The proof uses a fixed-point argument: firms' optimal decisions depend continuously on the anticipated distribution of future structures, and that distribution depends continuously on those decisions.

Interpretation. The paper closes the model. It is not only a collection of best responses; there is an actual industry equilibrium consistent with those best responses.

3.9 Result 4: ergodic industry dynamics

Once equilibrium exists, the industry structure follows a Markov chain on a finite state space. The paper proves:

- there is a unique positive recurrent communicating class $R \subseteq \mathcal{S}$,
- there is a unique invariant probability measure μ^* on industry structures,
- and from any initial structure, the distribution of industry states converges to μ^* .

Equivalently, the long-run fraction of time spent in any recurrent structure equals its invariant probability.

Interpretation. The industry does not settle at a single fixed structure. Instead, it keeps evolving forever, but in a statistically regular way. This is the paper’s deepest dynamic result.

3.10 Economic meaning of ergodicity

The paper emphasizes three implications of ergodicity.

- **No terminal structure.** All viable recurrent structures are visited infinitely often.
- **Long-run regularity.** Time averages converge to the invariant distribution, so there is a meaningful notion of a “natural” distribution of the number of incumbents, entrants, and exits.
- **Forgetting initial conditions.** The effect of the initial industry structure eventually disappears.

Interpretation. This lets the model speak to both short-run industry evolution and long-run average patterns.

3.11 Numerical example: homogeneous-product Cournot industry

The paper illustrates the theory with a homogeneous-product Cournot example. Firm i ’s output is q_i , total output is $Q = \sum_i q_i$, and inverse demand is

$$p(Q) = D - Q.$$

Firm-specific marginal cost is

$$\theta_\omega = \gamma e^{-\omega},$$

so a higher success state ω means a lower marginal cost. With fixed operating cost f , current profits are

$$A_i = p(Q)q_i - \theta_i q_i - f.$$

In the Cournot spot-market equilibrium,

$$q_i^* = \max\{0, p^* - \theta_i\}, \quad p^* = \frac{D + \sum_{j=1}^{n^*} \theta_j}{n^* + 1},$$

where n^* is the number of active producers.

Hence firm profits can be written as

$$A(\omega, s) = (p^*(s) - \theta_\omega)^2 - f,$$

for active firms.

The paper then imposes a simple innovation structure:

- research raises the efficiency index by one with probability $\frac{ax}{1+ax}$,
- otherwise the firm stays in place,
- and outside competition shifts the state downward by one with probability δ and leaves it unchanged with probability $1 - \delta$.

There is at most one entrant per period in the example. The authors compare three environments under the same primitives:

- **MPN:** the decentralized Markov-perfect Nash industry,
- **Colluder:** a multi-plant monopolist maximizing total profits,
- **Planner:** a social planner maximizing consumer surplus.

Interpretation. The example is not there to estimate anything. It is there to show what the model can generate once one feeds in concrete functional forms.

4 Identification and instruments (what makes the estimates credible)

4.1 What standard approaches miss

- **Representative-firm models miss turnover.** They cannot naturally generate simultaneous entry and exit or very different firm outcomes from similar starting points.
- **Deterministic dynamic models miss flux.** If industry structure converges to a fixed point, they cannot explain the continual structural churning seen in plant-level data.
- **Purely competitive models often make structure passive.** In Ericson and Pakes, structure is an equilibrium object jointly shaped by investment, entry, exit, and strategic interaction.

4.2 What assumptions are doing the work

- **Monotonicity of profits.** Higher ω raises profits; stronger competition lowers them.
- **Bounded investment returns.** Investment improves future states only probabilistically and with bounded support.
- **Outside competition.** A persistent exogenous negative drift prevents firms from living forever without continued successful investment.
- **Free entry with a setup lag.** Entry is endogenous but costly, which disciplines the number of entrants.

Interpretation. These are not econometric instruments. They are the structural restrictions that generate finite firm lives, finite industry size, and ergodic industry dynamics.

4.3 Why compactness is crucial

A major technical obstacle is that the industry could, in principle, grow without bound. The paper's Proposition 2 solves this by proving that enough competition drives firm value arbitrarily close to the exit value. This makes entry finite and the state space effectively compact.

Why it matters. Without compactness, neither the existence proof nor the computation of equilibrium would go through cleanly.

4.4 Why ergodicity is informative

The ergodic result is credible because the model produces a finite-state Markov process with a unique recurrent class. Once that is true, the model has a sharp long-run prediction: average industry outcomes can be summarized by one invariant distribution rather than by a knife-edge terminal structure.

Why it matters for empirical work. If data come from industries observed at different dates in their histories, the ergodic view provides a natural way to interpret long-run averages and cross-sectional structure without knowing initial conditions exactly.

4.5 What is left reduced form

- **Spot-market competition.** The function $A(\omega, s)$ is left general in the theory. The paper does not force one particular product-market model at the abstract level.
- **Beliefs about rivals.** Firms do not observe rivals' current investment choices directly. Instead, the theory summarizes the resulting uncertainty through the transition kernel for structures.
- **Entry process details.** The sequential free-entry rule is one natural closure, but other entry mechanisms could be inserted in later work.

4.6 Relation to the literature

- **Versus Jovanovic and Hopenhayn.** This paper allows both firm heterogeneity and strategic interaction, with endogenous investment affecting future states.
- **Versus dynamic duopoly papers.** It provides a more general finite-firm framework with endogenous entry and exit, rather than a specialized two-firm environment.
- **For later empirical IO.** The paper became a foundation for later work on dynamic oligopoly estimation and computational industry dynamics.

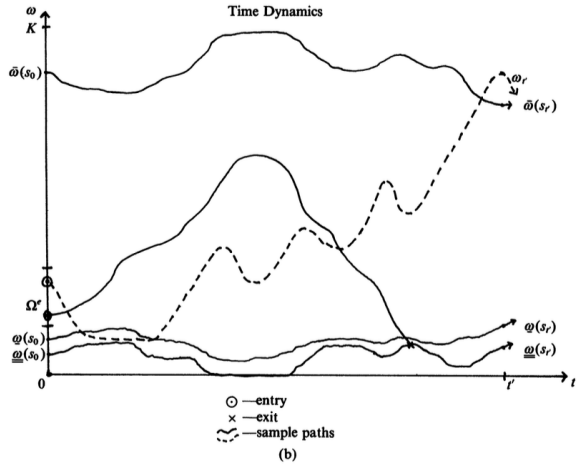
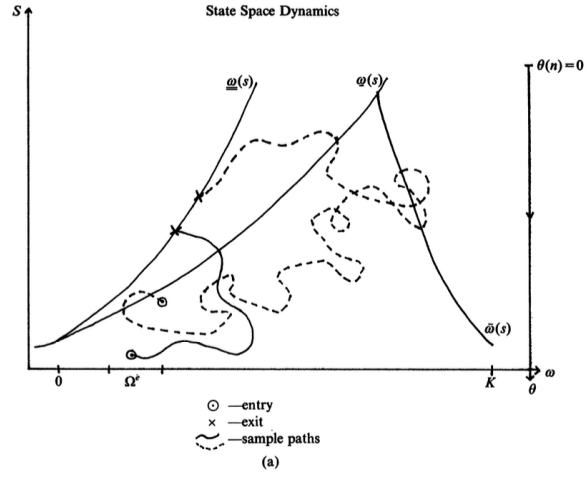


FIGURE 1

5 Tables and figures

5.1 Figure 1: State-space dynamics and time dynamics

Read:

- The top panel draws the state-space logic of the model. Firms enter in entry states Ω^e , invest to move upward, can coast without investing in a middle region, and eventually hit an exit boundary if their state deteriorates too much.
- The bottom panel translates the same logic into time paths for example firms. Similar entrants can have very different life histories because of idiosyncratic shocks.
- This figure is the paper’s visual summary of why the model can produce rank reversals, high mortality, and large heterogeneity in firm trajectories.

5.2 Table 1, panel A: Number of active firms and turnover rates

Read:

- In the MPN equilibrium, the industry is most often a duopoly: two firms are active in **70.8%** of simulated periods, while one firm is active in **27.9%**.
- Entry and exit are common under MPN: simultaneous entry and exit occurs in **16.5%** of periods, and at least one of the two occurs in **20.4%**.
- The colluder and planner produce much more concentrated industries. The colluder has one active firm in **92.4%** of periods, and the planner in **98.3%**.
- So decentralization generates both more firms and more churn.

5.3 Table 1, panel B: Prices, investment, entry, and number active

Read:

- Average price under MPN is **1.79**, well below the collusive benchmark of **2.22**. The planner’s price equals marginal cost, so the table does not report it in the same way.
- Total investment is highest under MPN at **1.05**, compared with **0.68** for the colluder and **0.84** for the planner.
- Average entry is also highest under MPN: **0.19** entrants per period, versus **0.08** for the colluder and **0.02** for the planner.
- Average number of active firms is **1.74** under MPN, but only **1.08** for the colluder and **1.02** for the planner.

5.4 Table 1, panel C: Welfare comparison

Read:

- Discounted welfare is highest for the planner, **58.8**, as expected.
- The MPN equilibrium delivers discounted welfare of **36.5**, which is higher than the colluder's **28.1** despite having a more fragmented and more turbulent market structure.
- A striking point emphasized by the authors is that similar-looking market structures need not have similar welfare properties, and very different structures can sometimes yield welfare levels that are surprisingly close.
- The standard deviations of welfare across simulation runs are large, which is a warning against overinterpreting any single realized sample path.

TABLE 1

<i>Simulated quantities from a homogeneous product model*</i>			
A. % of Simulated Periods with	MPN	Colluder	Planner
1 firm active	27.9	92.4	98.3
2 firms active	70.8	7.6	1.7
3 firms active	1.2	0	0
4 firms active	0.1	0	0
Entry and Exit	16.5	5.4	1.2
Entry or Exit	20.4	10.0	2.1
B. Average (standard deviation of)			
Price	1.79 (0.35)	2.22 (0.36)	**
Total investment	1.05 (0.41)	0.68 (0.29)	0.84 (0.41)
Entry	0.19	0.08	0.02
Number active	1.74 (0.48)	1.08 (0.27)	1.02 (0.13)
C. Welfare Runs (average and, in parenthesis, standard deviation of)			
1. Discounted Consumer Benefits	27.4 (6.4)	6.6 (5.5)	**
2. Discounted Net Cash Flow	11.6 (5.4)	22.5 (8.5)	—
3. Discounted Entry-Exit Fees	2.5 (1.0)	1.0 (1.0)	—
4. Discounted Welfare	36.5 (11.9)	28.1 (14.1)	58.8

* All runs are based on the specification described in the text with the following parameter values $D=4$, $f=0.2$, $x^*=0.4$, $w^0=4$, $\phi=0.2$, $c=1$, $S=0.7$, $a=3$, $\beta=0.925$. Panels A and B are obtained from a run which starts with one firm entering the industry, goes 10,000 periods, and then calculates the appropriate descriptive statistics. Panel C is obtained by doing 100 runs, each starting with one entrant and each lasting 100 periods. The appropriate discounted values are taken from each run, and then their averages and standard deviations across runs are computed.

** The welfare result for the planner can be read off the value function which is computed exactly. The planner sets price equal to the marginal cost of the minimum cost producer. This minimum marginal cost averaged 0.15 with a standard deviation of 0.39.

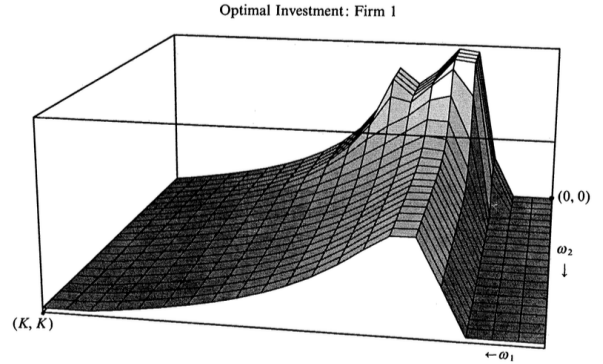


FIGURE 2

5.5 Figure 2: Optimal investment surface

Read:

- Investment is initially increasing and then decreasing in the firm's own success state. That is, successful firms invest more for a while, but eventually the value function flattens enough that extra investment is no longer worth it.
- In the example, investment starts around own state $\omega = 3$ or 4, depending on the competitor's state.
- The most interesting feature is the spike in investment at high own states. The paper interprets this as **entry deterrence**: once a firm is strong enough, it invests aggressively to keep potential entrants out.
- The figure also helps explain why entrant life cycles are very skewed: most entrants begin with low investment and die young, but the few that receive good early draws invest more and can survive for a long time.

6 Short summary

- Ericson and Pakes build a dynamic industry model in which firms invest to improve their state, face idiosyncratic success and failure, enter endogenously, and exit when continuation is no longer worthwhile.
- The model has strong structural results: unique optimal policies, finite entry, existence of Markov-perfect equilibrium, and ergodicity of the equilibrium industry process.
- In the example, decentralized competition generates more investment, more entry and exit, lower prices, and higher welfare than collusion, while still remaining far from the planner's allocation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that one can model industry evolution with heterogeneous firms, endogenous entry and exit, and idiosyncratic shocks in a way that is both theoretically coherent and useful for empirical work.
- It proves that the resulting equilibrium can be summarized as a Markov process over industry structures with a unique invariant distribution.
- It also shows that rich firm turnover and persistent industry flux are not pathological features of a model; they are equilibrium outcomes of optimization under uncertainty.

7.2 Economic intuition

- Firms are constantly running against two forces: their own investment can improve their competitive position, but outside competition and rivals' success keep eroding it.
- Because success is uncertain and investment is costly, some firms die quickly, some coast, and a few survive long enough to dominate temporarily.
- Entry keeps the industry alive, but competition keeps it from exploding. The long-run industry is therefore not a fixed point but a recurrent stochastic pattern.

7.3 Takeaway

This paper is best thought of as a foundational language for dynamic industrial organization. Its big message is that industry structure is itself an equilibrium random variable, shaped jointly by investment, entry, exit, and strategic interaction. Once we take that seriously, many facts that look messy in micro data—simultaneous entry and exit, persistent heterogeneity, and unstable market structure—become natural outcomes of a disciplined dynamic model.